

BatSpectroGen

User Manual | Workshop edition | 30 September 2026

BatSpectroGen processes folders of WAV recordings into a series of spectrogram images and, when selected, an average power-spectrum image for each recording. This version retains the familiar ChiroVerse interface while improving reliability and memory use on a wider range of computers.

The screenshot shows the BatSpectroGen interface with the following elements:

- Input Folder:** A text box with a "Browse" button.
- Output Folder (optional):** A text box with a "Browse" button.
- Segment Duration (s):** A text box containing "5".
- Number of Threads:** A text box containing "1".
- Processing Profile:** A dropdown menu showing "balanced".
- Colormap:** A dropdown menu showing "magma".
- Generate Power Spectrum:** A checked checkbox.
- Use Multiprocessing:** An unchecked checkbox.
- Filter (Select frequencies to display):** A section with "Min (kHz)" and "Max (kHz)" text boxes, and an "Apply filter to Power Spectrum" checkbox.
- Buttons:** A green "Generate Spectrograms" button and a red "Stop" button.
- Progress Bar:** A bar showing "0% | 0/0 files".
- Log:** A large text area for logging.
- About:** A blue button in the bottom right corner.

Interface note: BatSpectroGen reads the sampling rate directly from each WAV file. Select a **Processing Profile** to control image resolution and memory use.

1. Before you begin

- Use Windows 10/11, a current macOS release, or a modern Linux distribution.
- Install 64-bit Python 3.9, 3.10, 3.11, 3.12 or 3.13. Windows users should select **Add Python to PATH**.
- Keep the entire workshop folder together after downloading and extracting it.
- Allow enough free space for the WAV files and generated JPEG images; 4 GB free space is a practical minimum for the workshop.

2. Installation

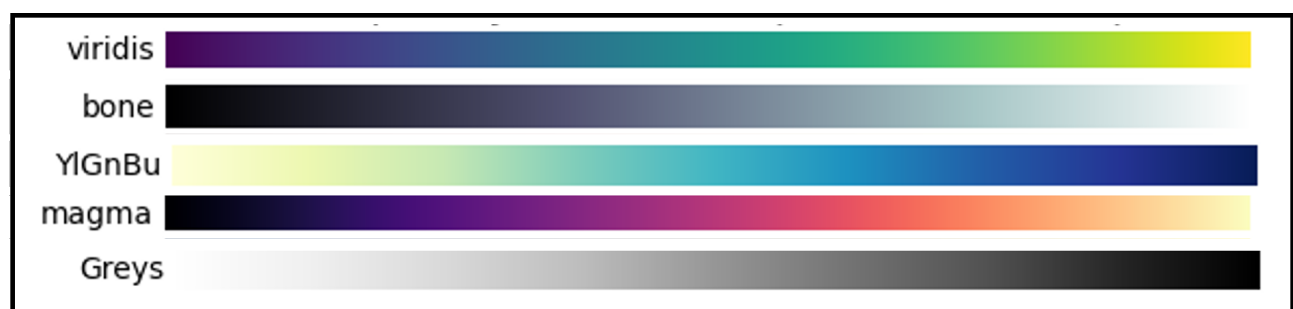
Windows	Double-click Install_BatSpectroGen_Windows.bat once. When setup finishes, start the program with Run_BatSpectroGen_Windows.bat .
macOS	In Terminal, open the BatSpectroGen folder and run: <code>chmod +x Install_BatSpectroGen_macOS_Linux.sh</code> <code>Run_BatSpectroGen_macOS_Linux.sh</code> <code>./Install_BatSpectroGen_macOS_Linux.sh</code> <code>./Run_BatSpectroGen_macOS_Linux.sh</code>
Linux	Install Python, Tk and libsndfile first. On Ubuntu/Debian: <code>sudo apt install python3 python3-venv python3-tk libsndfile1</code> Then run the same setup and launch commands shown for macOS.

3. GUI controls

Input Folder	Select the folder that directly contains the WAV recordings.
Output Folder (optional)	Choose a new or empty folder. If left blank, BatSpectroGen creates a BatSpectroGen_Output_... folder beside the input folder.
Segment Duration (s)	Length represented by each image. Use 5 seconds for the workshop unless instructed otherwise.
Number of Threads	Used only when Use Multiprocessing is selected. Start with 1.
Processing Profile	Select Low-memory, Balanced or High-detail. See the table below.
Colormap	Select the colour scale used for the spectrogram.
Generate Power Spectrum	Creates one average relative power-spectrum image for each recording.
Use Multiprocessing	Processes multiple recordings concurrently. Leave this off on low-power or low-memory computers.
Min / Max (kHz)	Optional displayed frequency range. These values change the plot only; the audio is not filtered or modified.
Apply filter to Power Spectrum	Uses the same displayed frequency range for the power-spectrum image.

4. Processing profiles

Profile	Output	When to use it
Low-memory	1600 × 900 smaller FFT	Older laptops, 4 GB RAM, or when the computer becomes slow. Power spectrum is off by default.
Balanced	1920 × 1080	Recommended for the workshop and most survey workflows.
High-detail	3840 × 2160 larger FFT	Use when the additional image and frequency resolution is needed and sufficient time, memory and disk space are available.



Available colormap examples. Choose one consistently when comparing recordings.

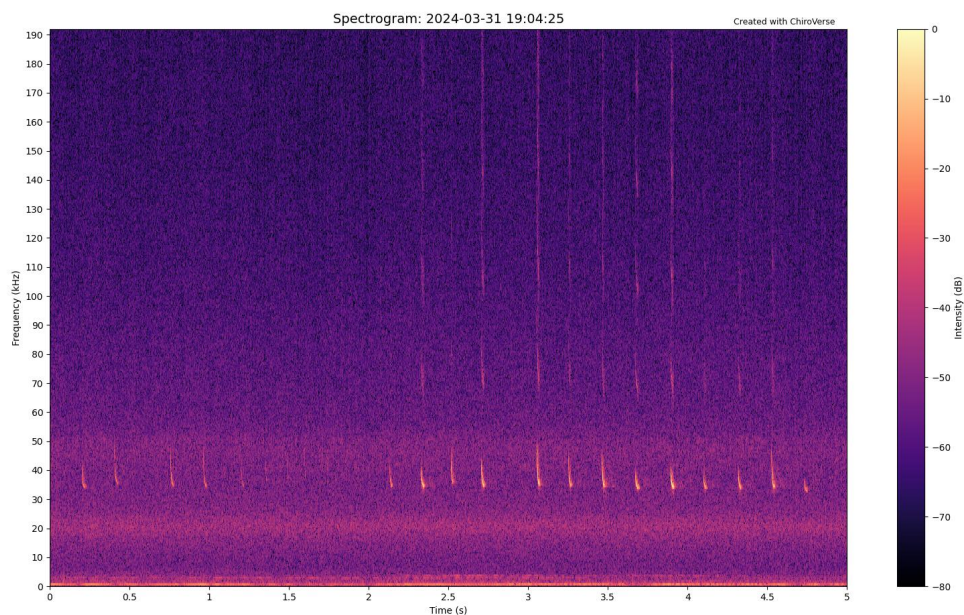
5. Workshop workflow

1. Place the WAV recordings for one batch in a clearly named folder.
2. Start BatSpectroGen using the launch file for your operating system.
3. Select the Input Folder and a new or empty Output Folder.
4. Begin with Segment Duration = 5, Profile = Balanced, one thread, and Use Multiprocessing switched off.
5. Select a colormap and, if needed, enter the displayed Min and Max frequencies in kHz.
6. Select Generate Spectrograms and monitor the progress bar and log.
7. When processing finishes, inspect several images before using the complete batch.

Important: the output folder must be empty. This prevents outputs made with different settings from being mixed. The Stop button requests a safe stop; current work may finish before the program stops.

6. Understanding the output

- Each WAV recording receives a subfolder named after the source file.
- Spectrogram JPEGs are timestamped and numbered in recording order.
- A shorter final segment is retained when the recording length is not exactly divisible by the selected duration.
- The optional average power spectrum summarizes the whole recording without loading the complete file into memory.
- The output contains the generated images only; no CSV manifest or JSON run-summary file is created.



Example spectrogram output. The frequency axis is in kHz and the colour bar shows relative amplitude in dB.

7. Performance and reliability

- For a first run, use Balanced, one thread and multiprocessing off.
- On older or low-memory computers, choose Low-memory and turn off the power spectrum if it is not required.
- On an 8 GB computer, try two threads only after a one-thread test succeeds. More threads use more memory and may not be faster.
- Use High-detail only when the extra 3840×2160 output is useful; it needs more time and disk space.
- Close unnecessary applications and avoid writing directly into a synchronised cloud folder while processing.
- BatSpectroGen reads one segment at a time, reads each WAV file's actual sampling rate, and does not silently resample the recording.

8. Troubleshooting

Program does not open: Run the setup file first. On Windows, confirm that Python was added to PATH.

No WAV files found: Select the folder that directly contains the .wav files.

Output folder is not empty: Choose a new folder or move the previous results elsewhere.

Invalid frequency range: Min must be lower than Max. Max cannot exceed half the WAV sampling rate.

Computer becomes slow: Choose Low-memory, switch multiprocessing off, use one thread, and close other applications.

macOS blocks the script: Right-click and choose Open, or allow it in Privacy & Security settings.

Linux reports a Tk error: Install the Tk package for the active Python version, for example python3-tk.

If a run fails, keep the message shown in the log and note the source WAV filename. These make the problem easier to diagnose.

9. Credits and support

Created by **Vedant Barje** and **Kadambari Deshpande**, ChiroVerse.

Supported by the Indian Institute for Human Settlements, Bengaluru, and Wildlife Conservation Trust, Mumbai.

Contact: connect.chiroverse@gmail.com

Repository: github.com/ChiroVerse/BatSpectroGen_ChiroVerse

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